

Specifically, we focus on how well they can interpret multimodal inputs, form generalizations from observations, and apply the general principles to solve specific cases. Furthermore, they may reason differently about fundamental concepts such as numbers, colors, shapes, and size. Hence, we propose PUZZLEVQA, a diverse collection of abstract pattern puzzles to diagnose reasoning challenges in multimodal models. The dataset is automatically constructed through multimodal templates, and includes reasoning explanations for interpretability.

3.1 Puzzle Components

As shown in Figure 2, each puzzle in our dataset is formulated with the following main components:

1. **Objects:** The conceptual elements that interact within the puzzle, such as numbers, colors, shapes, and size.
2. **Layout:** The spatial arrangement of objects that provides visual context.
3. **Pattern:** The relationship that governs the interaction amongst objects. For example, a pattern may be that spatially opposite parts must have the same color.
4. **Demonstrations:** Multiple instances of interacting objects that collectively represent the underlying pattern. Without demonstrations, the pattern would become ambiguous.
5. **Query:** The natural language question that directs the multimodal model how to solve the puzzle by determining the missing object.

3.2 Design Considerations

We have three main design considerations for each puzzle in our dataset:

Simplicity. As the focus is on evaluating how large multimodal models reason about fundamental abstract concepts, we do not deliberately make the puzzles more complex than necessary. We also aim to make the underlying patterns straightforward, without requiring extensive world knowledge or advanced theories.

Correctness. To avoid potentially noisy annotations, we use an automatic approach with multimodal templates to generate each puzzle. For instance, given a visual layout and pattern of a template, we can automatically populate the template

with the appropriate objects that demonstrate the pattern. As each puzzle instance is created based on the specific rules in the template, we can ensure that they do not contain annotation mistakes.

Diversity. To investigate the multimodal reasoning capabilities across diverse abstract concepts, we construct puzzles based on four main concepts: numbers, colors, shapes, and size. Furthermore, to evaluate how well the models can relate to multiple concepts, we design both single-concept and dual-concept puzzles.

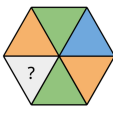
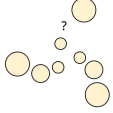
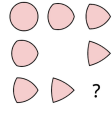
3.3 Puzzle Construction

Multimodal Templates. To construct each abstract puzzle, we leverage multimodal templates based on fundamental concepts including numbers, colors, shapes, and size. Following the formulation and design considerations previously discussed, we first define the layout and abstract pattern for the puzzle. Each template can be randomly populated with the specific objects to represent the underlying pattern through demonstrations, forming a specific puzzle instance. For example, to construct a color-based puzzle instance shown in Figure 2, we focus on the concept of colors and define the layout as a hexagonal arrangement of six parts, with the abstract pattern that spatially opposite parts must have the same color. Thereafter, the template can be randomly populated to satisfy the pattern with colors from a predefined list of possible colors. Lastly, the query is constructed based on the fundamental concepts in the abstract pattern. To demonstrate our puzzle generation pipeline, we include a detailed implementation in Appendix A.4, based on the puzzle in Figure 1.

Reasoning Explanations. To ensure that each abstract puzzle can be easily understood, we also construct reasoning explanations based on the three problem solving stages: image descriptions for visual perception, pattern explanations for inductive reasoning, and deductive reasoning steps. Specifically, we leverage textual templates that can be populated with details from the specific puzzle instance, as shown in Figure 2. In our experiments in Section 5, this enables us to identify reasoning bottlenecks by progressively providing the explanations of each stage to the model.

3.4 Multiple-Choice Format

While we use straightforward objects in our puzzles, there may be a degree of ambiguity in the an-

Numbers	Colors	Size	Shapes
			
What is the missing number of the part denoted with a question mark?	What is the missing color of the part denoted with a question mark?	What is the size of the missing circle denoted with a question mark?	What is the missing shape denoted with a question mark?
(A) 2 (B) 4 (C) 1 (D) 6	(A) red (B) blue (C) green (D) yellow	(A) small (B) large (C) medium	(A) triangle (B) circle (C) hexagon (D) square

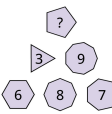
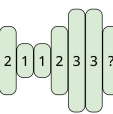
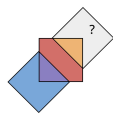
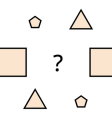
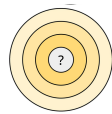
Numbers & Shapes	Numbers & Colors	Numbers & Size	Shapes & Colors	Shapes & Size	Colors & Size
					
What is the missing number of the part denoted with a question mark?	What is the missing number of the part denoted with a question mark?	What is the missing number of the part denoted with a question mark?	What is the missing color of the part denoted with a question mark?	What is the size of the missing shape denoted with a question mark if it is a square?	What is the missing color of the part denoted with a question mark?
(A) 5 (B) 7 (C) 9 (D) 2	(A) 8 (B) 3 (C) 1 (D) 6	(A) 1 (B) 2 (C) 3 (D) 4	(A) red (B) blue (C) green (D) yellow	(A) large (B) medium (C) small	(A) light yellow (B) light blue (C) dark yellow (D) dark red

Figure 3: Taxonomy of abstract puzzles in PUZZLEVQA with sample questions, based on fundamental concepts such as colors and size. To enhance diversity, we introduce both single-concept and dual-concept puzzles.

swer regarding specific colors or sizes. Hence, we standardize the puzzle format as multiple-choice questions, where all questions are provided with four options, with the exception of three options for size (small, medium, and large). To generate the incorrect choices for each question, we use heuristics including randomly sampling numbers within the same magnitude of the answer, and further details can be found in the supplementary material. We use the standard accuracy metric for evaluation.

3.5 Dataset Analysis

To ensure that the dataset contains diverse abstract patterns, we provide a taxonomy of 10 puzzle categories based on fundamental concepts including numbers, colors, shapes, and size. As shown in Figure 3, there are four categories of single-concept patterns. To extend the depth of PUZZLEVQA, we also include dual-concept patterns, which would require models to relate two concepts in order to solve the puzzle. Within each category, we design two multimodal templates that can each be used

to generate many unique puzzle instances. The full list of puzzle templates and examples of more puzzle instances can be found in the supplementary material. To maintain a reasonable dataset size for evaluating large multimodal models, we generate 100 unique puzzle instances from each template. Thus, there are 2000 test instances in PUZZLEVQA in total. We conducted an analysis in Appendix A.5 and found that the chosen dataset size is large enough to be relatively robust to experimental variance.

3.6 Implementation Details

We utilize Python code along with packages like Pillow² to automatically generate puzzles. Leveraging these tools, we are able to create many different unique puzzle images and text questions for each given puzzle type by augmenting the base template and objects in the image. Example code snippets to generate the puzzles are included in the supplement-

²<https://pypi.org/project/pillow/>

	Numbers	Colors	Size	Shapes	Average
Random Baseline	25.0	25.0	33.3	25.0	27.1
Qwen-VL-Chat	25.0	22.0	32.5	34.0	28.4
LLaVA-13B	26.0	29.0	29.0	26.0	27.5
Gemini Pro	32.5	32.0	33.5	40.0	34.5
Claude 3 Opus	47.0	32.5	33.5	44.5	39.4
GPT-4V	67.5	42.0	35.0	41.0	46.4

Table 1: Accuracy of large multimodal models for single-concept abstract patterns in PUZZLEVQA.

tary material, and we plan to release the dataset publicly with a permissive license such as MIT license.

Each puzzle in PUZZLEVQA comprises of an image x_{image} , a natural language question $x_{question}$, an image caption that describes the image $x_{caption}$, an explanation that explains the pattern shown in the image $x_{explanation}$, a deduction statement that applies the pattern to the puzzle to derive the final answer $x_{deduction}$, a set of multiple-choice answers $x_{options}$, and the final answer x_{answer} . All of which are automatically generated during the puzzle creation process.

4 Experimental Setup

4.1 Inference Pipeline

To elicit reasoning steps from large multimodal models, we leverage zero-shot chain of thought (CoT) prompting (Kojima et al., 2022) with a prompt similar to “Let’s think step by step”. As the model may not always follow the same multiple-choice answer format, we also employ a model-based answer extraction stage. Detailed examples of the prompts can be found in the supplementary material. Please note that our main experimental setting used in Table 1 and 2 involves only the questions and images as multimodal inputs. On the other hand, we progressively provide additional ground-truth information such as image captions in Section 5.1 to diagnose the multimodal reasoning bottlenecks.

Chain of Thought Prompting. In the first prompting step, we construct a prompt $\hat{x}$ by modifying the question using a specific prompt template: “[I] Question: [X]. Options: [O]. Answer: [T]”, where [I] is the input slot for x_{image} , [X] is the input slot for $x_{question}$, [O] is the input slot for $x_{options}$, and [T] is the input slot for the trigger sentence t_1 . To elicit reasoning over the multimodal inputs, we use “Let’s describe the image first and think step by step” as our trigger sentence

t_1 . This modified prompt $\hat{x}$ is then fed into a large multimodal model, and a greedy decoding strategy is utilized to generate the subsequent sentence y_1 . If the letter answer can be extracted from y_1 with regular expressions, the prompting process terminates. However, if the letter answer cannot be extracted, we prompt the model itself to extract the answer.

Answer Extraction. In the second prompting stage, we use the generated sentence y_1 along with the modified prompt $\hat{x}$ to extract the final answer. We concatenate three elements to form “[$\hat{X}$] [Z] [A]” where [$\hat{X}$] is the input slot for $\hat{x}$, [Z] is the input slot for y_1 , and [A] is the trigger sentence t_2 to extract the final answer. We defined t_2 as “Therefore, among (A) (B) (C) (D), the answer is:” or “Therefore, among (A) (B) (C), the answer is:”, for puzzles with four and three multiple-choice questions respectively.

4.2 Models

To investigate the reasoning ability of large multimodal models, we select the best-performing open and closed-source models (Yue et al., 2023):

1. Qwen-VL-Chat (7B) (Bai et al., 2024) is an open-source large multimodal model designed to perceive and understand both texts and images. We use the version with open model weights and default chat template.
2. LLaVA-13B (Liu et al., 2023) is a large multimodal model which is based on the popular LLaMA (Touvron et al., 2023a) foundation language model. We use the model weights of the 1.5 version and default chat template.
3. Gemini Pro (Gemini Team, 2023) is a highly capable multimodal model released by Google, and we use their publicly available API to query the “gemini-pro-vision” version of the model.
4. Claude 3 Opus³ is released by Anthropic and the most highest-performing multimodal model in their model family. We use their publicly available API to query the “claude-3-opus-20240229” version of the model.
5. GPT-4V (OpenAI, 2023) is released by OpenAI and widely regarded as the most capable

³<https://www.anthropic.com/news/claude-3-family>